2.6 Selecting signals

When collecting signals, be sure to consider the following things:

- Why should I capture this signal? From an ROI perspective, does the signal yield an actionable insight, or does it contribute to the noise (and, even worse, potentially cause network and storage-related costs, without offering any benefit)?
- How long should I store the signal (retention period)? This depends chiefly on the task at hand (e.g., troubleshooting, on-call, or optimization), but there may also be external factors that influence the decision on how long to retain a signal. For example, in the financial industry, you may be required to keep transaction logs for a certain period of time, in order to be, say, Payment Card Industry Data Security Standard (PCI DSS) compliant.
- For numerical signals, such as metrics, what is the "normal" value (or value range)? This is usually used as a baseline. You may consider automating the process of establishing the baseline, potentially using machine learning techniques. In this context, if a signal is not in its normal range, anomaly detection (https://github.com/AICoE/prometheus-anomaly-detector) is your friend.
- Beware of signal hoarding; keeping around all metrics or logs, just because you may need them one day (without a strategy), can be expensive and negatively influence the ROI.
- Label consistently across signal types (e.g., Loki, Prometheus, and Parca).

With this general collection of good practices and considerations regarding selecting signals, we have reached the end of the chapter.

Summary

- To generate actionable insights, for observability, we rely on different telemetry signals, most importantly logs, metrics, and traces.
- You need to decide what signals to collect, store, and use, based on their utility and costs.
- Logs are signals with a textual payload that capture an event, usually consumed by humans.
- Ideally, logs should be structured (e.g., expressed in JSON) and come with a schema, but in reality, the situation can be messy, due to the many formats and protocols in use.
- Metrics are numerical signals, useful for automation (e.g., for dashboarding and alerting).
- The Prometheus format has established itself as a standard in cloud-native systems.
- Distributed traces are signals that capture and visualize how a request moves through your microservices, showing service duration and success state.